Stability of the Magnetization of the Hurley Creek Formation

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Oriented samples were taken from five sites in the Hurley Creek formation of New Brunswick. This formation lies immediately over the Minto formation, which is well-dated by fossils. Thermal cleaning shows that only one very stable magnetization is present. The between-site dispersion is consistent with a magnetization acquired during or shortly after deposition. The mean directions ($D = 171°$E and $I = +09°$) of the Hurley Creek formation are therefore considered representative of the Pennsylvanian field of New Brunswick, Canada. The corresponding pole position is 39°N, 125°E.

Introduction. In the past decade, paleomagnetic results from a few Upper Carboniferous formations have been obtained from the eastern section of North America [*Dubois*, 1959; *Roy*, 1963, 1966; *Black*, 1964]. Most of the formations sampled are Pennsylvanian from the Maritime provinces of Canada where complex structure makes correlation difficult. *Gussow* [1953] uses coal deposits to correlate the formations within the Pennsylvanian of the Maritimes. He chooses the Minto district as a type area, because it is well known from the extensive coal workings.

The geology of the Minto-Chipman area has been described by *Muller* [1951]. In his work, the Pictou group is subdivided into the Minto (lower and upper member), the Hurley Creek, and the Sunbury Creek (youngest) formations. The age of the Minto coal seam (upper member of the Minto formation) has been determined from collections of fossils made by *Dyer* [1926] and *Muller* [1951] and identified by W. A. Bell. A Westphalian C age was established by Bell for Muller's collection. Accordingly, *Gussow* [1953] places the Pictou group in the upper part of the Pennsylvanian. Because of their dating, it seemed worth while to obtain some paleomagnetic results from these formations.

Sampling. Natural outcrops suitable for paleomagnetic purposes could not be found in the Minto area. However, coal deposits south of Minto are strip-mined, exposing the Hurley Creek and the Minto formations. Unfortunately, these trenches are back-filled, so that it is impossible to revisit a site. Three surveys (May 1965, May 1966, August 1966) were made in order to sample at three fresh digging sites; two partly filled old sites were also sampled, giving five sites in all. Collecting details are given in the appendix. Location of the sites are shown in Figure 1. Burial of sites by backfilling makes it difficult to correlate the respective levels of the sites; however, it is estimated that a thickness of 5 to 10 meters was sampled. All samples were taken from the Hurley Creek formation, which is composed of red to purplish red shales, sandstones, and siltstones. The shales were not sampled. Drilling was impossible because of the difficult access to the sites and hand samples were taken instead. The Minto formation was not found suitable for paleomagnetic sampling. Two specimens (cylindrical cores of 2.5-cm diameter and 2.2-cm height) were taken from each sample, except in two cases where only one specimen could be obtained from the small sample. The directions and intensities were measured by means of an astatic magnetometer. The orientation error in marking and transferring marks is estimated at 2° to 3°. The direction errors in measurements are less than 1° because of the large intensities of the specimens, and the intensity errors are about 3%.

Demagnetization. A specimen from each sample was thermally demagnetized by successively heating to higher temperatures and cooling in a zero (± 50 γ) field. The results are given in Table 1 and Figure 2. The mean directions D and I (Table 1) do not change significantly after each thermal cleaning even up to 650°C. The circular standard deviation (θ_{63}) after each cleaning varies very little and indi-

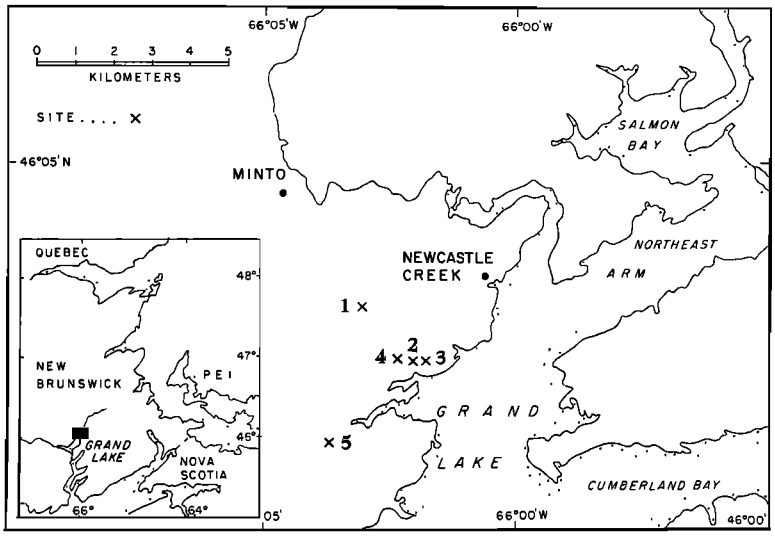


Fig. 1. Locality map showing sampling sites.

cates that the dispersion of the directions within a site is not affected by thermal cleaning. (Site 4 ($N = 1$) shows the behavior of an individual specimen.) It is therefore concluded that only one magnetization is present and that it is very stable. It is shown in Figure 2 that this magnetization has a high resistance to thermal cleaning. Only 10% of the blocking temperatures are below 375°C and 53% (60% when all samples are taken) are above 650°C. It is quite clear, in this case, that the carrier of magnetization is hematite and not magnetite, since there is no abrupt intensity decrease in the magnetite blocking temperature range. Examination of thin sections from site 1 confirmed that, in both the siltstone and the sandstone, the opaque minerals consisted of small hematite crystals and the angular fragments were embedded in a hematite cement.

The stable Hurley Creek rocks gave an ideal opportunity to compare on-center and off-center methods of measurement, using an astatic mag-

TABLE 1. Mean Site Directions and Within-Site Dispersion at Various Temperatures

Temperature, °C	Site 1 $N = 5$			Site 2 $N = 2$		Site 3 $N = 5$			Site 4 $N = 1$		Site 5 $N = 6$		
	D, deg	I, deg	θ_{63}	D, deg	I, deg	D, deg	I, deg	θ_{63}	D, deg	I, deg	D, deg	I, deg	θ_{63}
20	173	+05	6.8	185	+10	166	+04	9.0	158	+14	172	+19	10.1
125	172	+06	7.6	185	+09	168	+04	11.6	159	+11	173	+14	9.6
250	172	+06	6.6	187	+09	167	+03	11.2	159	+11	173	+12	8.9
375	172	+06	7.9	185	+09	166	+02	9.2	160	+12	173	+12	10.5
450	173	+06	6.6	186	+09	166	+04	9.4	160	+12	172	+15	8.8
525	172	+08	6.6	185	+09	166	+04	9.2	159	+15	168	+17	11.6
600	173	+07	7.0	186	+10	165	+03	9.5	159	+14	171	+18	8.6
650	175	+07	8.1	188	+09	165	+03	9.4	160	+12	169	+19	11.6

Notes

N is the number of samples per site.

θ_{63} is the circular standard deviation. θ_{63} not calculated for sites 2 and 4 because of insufficient data.

D is the mean declination east of true north.

I is the inclination positive downward.

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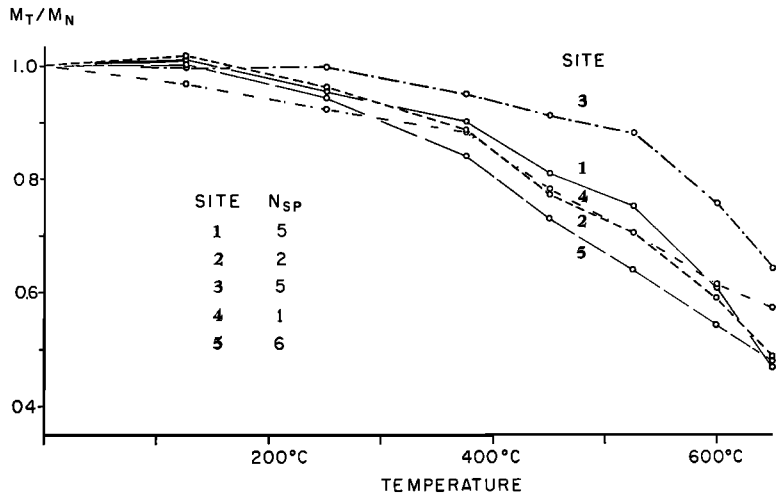


Fig. 2. Thermal decay of intensity normalized to unity at room temperature. N_{sp} is the number of specimens contributing to each curve.

netometer. (See *Collinson and Creer* [1960] for the description of the methods.) The off-center method, in which the specimen is placed below and off the axis of the magnetometer, has been criticized, because the field produced at the magnetometer by one component decreases as the fourth power of the distance whereas the field produced by the other component decreases as the third power. In the work described here, all NRM measurements have been made using the off-center method; all other measurements (after cleaning) have been made using the on-center (below and on the axis) method, in which the field of each component decreases as the same (third) power. No systematic difference in direction or intensity could be found between the NRM measurements and the measurements after cleaning. In fact, most of the individual observations (direction and intensity) before and after cleaning at 125°C were in close agreement, indicating that the off-center method did not introduce any detectable error.

All specimens were thermally cleaned at a temperature of 600°C. The mean directions (Table 2) of a sample were obtained by taking the mean of the direction cosines of each specimen. Then the site means were obtained by giving each sample unit weight. The sample directions are given in Figure 3, where a between-site dispersion may be observed. An F-distribution test [*Watson and Irving*, 1957] gives a value of $F = 4.5$, which is greater than the value at the 5% significance level ($F_{.05}$ (8, 28) = 2.3), and therefore confirms that the polar vectors at each site are different. The within-site and between-site estimates of precision are, respectively, 95 and 97, and the circle of confidence using the two-tier analysis of *Watson and Irving* [1957] is 7°.

Discussion. These observations show that the Hurley Creek formation has acquired a very stable magnetization and has been unaffected by any secondary magnetization. The difference between the Hurley Creek mean directions and the mean directions at that locality ($D = 005°$, $I = 31°$) obtained from North-American Triassic results (computed by *Irving* [1964]) is 14° in D and 40° in I. This is much greater than the statistical errors and therefore places the time of magnetization as pre-Triassic, i.e., Pennsylvanian or Permian. The pole positions given by North American Pennsylvanian and Permian formations show no evidence of polar wandering during this period. However, those results have been obtained from sedimentary formations exclusively. Although they indicate a direction prior to Triassic age, it has not been shown if the magnetization has been acquired during or shortly after deposition. The Pennsylvanian formations studied [*Runcorn*, 1956; *Dubois*, 1959; *Collinson and Runcorn*, 1960; *Roy*, 1963, 1966; *Black*, 1964] could, therefore, have possibly acquired their magnetization

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TABLE 2. Analysis of Directions before (M_n) and after (M_{600}) Cleaning

Site	N_{sp}	N_{sa}	M_n M_i	M_n D, deg	M_n I, deg	M_n R	M_n k	M_{600} M_i	M_{600} D, deg	M_{600} I, deg	M_{600} R	M_{600} k	Pole Position Lat °N	Pole Position Long °E
1	9	5	75	174	+05	4.97	119	45	174	+06	4.97	118	41	122
2	4	2	46	184	+10	2.00	*id*	25	185	+06	1.99	*id*	41	107
3	10	5	96	166	+04	4 96	89	69	166	+04	4.96	101	41	133
4	2	1	37	166	+13	1.00	*id*	24	161	+14	1.00	*id*	34	137
5	11	6	15	171	+17	5.90	48	08	171	+17	5.94	080	35	124
Group		5*	45†	172	+10	4 95	85	27†	171	+09(10°)‡	4.93	058	39	125(5°, 10°)§

Notes

N_{sp} = number of specimens.
N_{sa} = number of samples.
id = insufficient data for significant value.
* Number of sites.
† Logarithm mean of the site means.
‡ Circle of confidence ($P = 0.05$).
§ Semi axis (δ_p, δ_m) of the ellipse of confidence ($P = 0.05$).
R = resultant vector giving each sample unit weight.
M_i = intensity $\times 10^{-6}$ (emu system).
D, I have the same significance as in Table 1.
k is *Fisher*'s [1953] best estimate of precision.

during Permian time, which might account for the similarity of the Pennsylvanian and Permian directions. Consequently, previous results cannot help in determining whether the magnetization of the Hurley Creek formation occurred during Pennsylvanian *or* Permian time.

If, however, the between-site dispersion observed in the Hurley Creek formation is an effect of the secular variation of the field, we may have an indication of the time of magnetization. Indeed, in order to get an effect of secular variation, the various sites must have

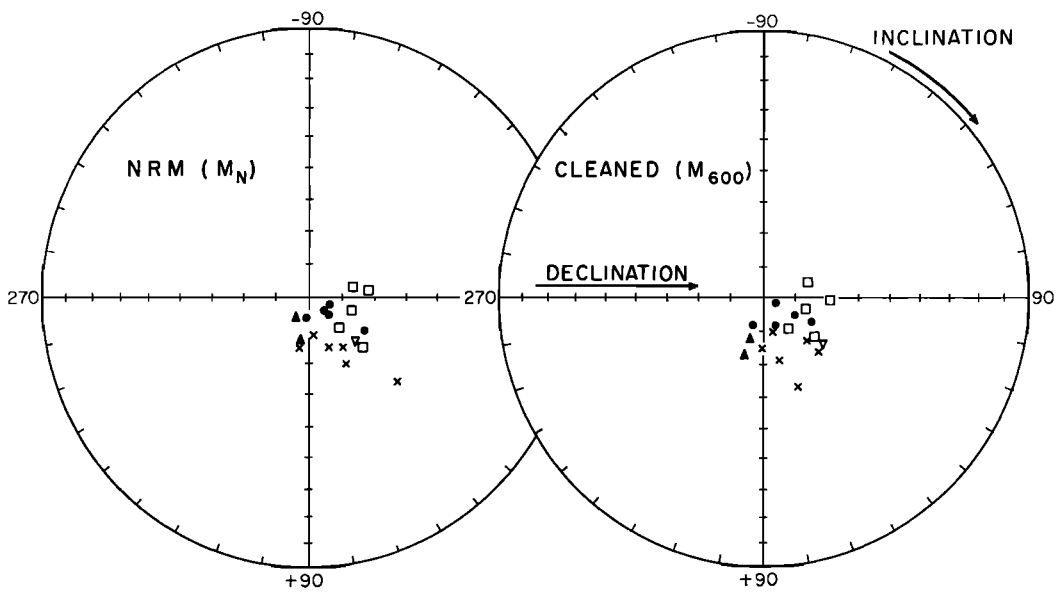


Fig. 3. Directions of magnetization before and after thermal cleaning. Equal area equatorial net. All directions between 90° and 270°. Solid circle, site 1; solid triangle, site 2; square, site 3; inverted triangle, site 4; cross, site 5.

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been magnetized at different times, i.e., during deposition or consolidation; otherwise, if the magnetization occurred after consolidation of all sites, say in Permian time, there would probably not be any effect of secular variation. Then, unless this between-site dispersion could be attributed to some other cause, it seems evident that the magnetization of the Hurley Creek formation is detrital or chemical and has been produced by a Pennsylvanian field.

The standard deviation of the site means is 11° and therefore smaller than the 18° calculated by *Creer* [1955] as the direction dispersion of the present field at the same latitude. This difference could indicate that the secular variation in Pennsylvanian time was smaller than the secular variation derived from *Creer's* [1955] analysis or that the time spanned by sampling is shorter than the period of secular variation. It is difficult to know the exact thickness sampled because only a small portion of the beds was uncovered at one time and then buried again. Furthermore, the rate of deposition of these sediments is not known.

The pole position (39°N, 125°E) of the Hurley Creek formation is within 8° of the pole position (41°N, 132°E) obtained from a second formation of the Pictou group [*Roy*, 1966]. The difference is not necessarily significant, since the 95% ellipses of confidence intersect, the semi axes σ_p and σ_m being 5°, 10° and 2°, 4°, respectively. On the other hand, this difference could be an effect of secular variation if either collection (or both) covers only part of the secular variation cycle. In fact, it is doubtful that the secular variation was averaged out in the results obtained from the second [*Roy*, 1966] formation. The standard deviation in that case was only 8°, and it was then found preferable to do the analysis on the over-all results of three formations in order to assure a large enough time span. Therefore, assuming that the difference between these results is significant, the mean direction of the Hurley Creek formation is considered more representative of a Pennsylvanian dipole field than that of the second [*Roy*, 1966] formation.

It is hoped in the near future to compare these results and their precision with results from igneous rocks of Pennsylvanian age, which presumably acquired a thermoremanent magnetization at the time of cooling.

Appendix. Reference List of Collecting Localities

All beds are flat-lying except at site 1 where they dip 03° at 358° of true north.

Site 1. Along Avon Co. opencut 3.5 km southeast of Minto. All samples from hard red sandstones. The thickness sampled is 8 meters starting about 9 meters above the top of the coal seam (Upper Minto formation).

Site 2. Southwest (3.3 km) of Newcastle Creek. All samples from relatively hard fine red siltstones; 5 meters below ground and 16 meters above coal seam.

Site 3. All samples from hard packed purple fine siltstones 300 meters east of site 2 and at same depth.

Site 4. Sample is similar to samples of site 2 but slightly coarser grained. Four hundred meters west of site 2.

Site 5. South (7 km) of Minto. All samples are of a lighter red color and of a larger grain size than samples from sites 1 to 4. The bed sampled is more than 13 meters above the coal seam and almost immediately above a thick grey bed. All samples within 1 meter stratigraphically.

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